INVERSE KAZHDAN–LUSZTIG POLYNOMIALS OF MATROIDS UNDER DELETION

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Abstract. We give a "deletion formula" for the inverse Kazhdan–Lusztig polynomial of a matroid: it relates the inverse Kazhdan–Lusztig polynomial of a matroid to that of a single non-coloop element deletion, and it is analogous to the deletion formula of Braden–Vysogorets for Kazhdan–Lusztig polynomials. As an application of our deletion formula, we produce a counterexample to a recent conjecture of Xie and Zhang concerning the real-rootedness of the Hadamard product of the inverse Kazhdan–Lusztig polynomial with a binomial. Furthermore, we ...

1. Introduction

The Kazhdan–Lusztig polynomial $P_{\mathsf{M}}(t)$ is a fundamental invariant associated with any matroid M. It was introduced by Elias, Proudfoot, and Wakefield in [EPW16]. This polynomial, denoted $P_{\mathsf{M}}(t)$, exhibits formal similarities to the Kazhdan–Lusztig polynomials defined for Bruhat intervals of Coxeter groups. The coefficients of $P_{\mathsf{M}}(t)$ depend only on the lattice of flats $\mathcal{L}(\mathsf{M})$ of the matroid and, in fact, they can be expressed as signed integer combinations of the flag Whitney numbers counting chains of flats with specified ranks (see [PXY18]).

The Kazhdan–Lusztig polynomial of M plays a significant role in the *singular* Hodge theory of matroids developed by Braden, Huh, Matherne, Proudfoot, and Wang in [BHM+22]. Two further polynomial invariants that are often studied in this context are the *inverse Kazhdan–Lusztig polynomial of a matroid*, introduced by Gao and Xie [GX21] and denoted $Q_{\mathsf{M}}(x)$, and the Z-polynomial introduced by Proudfoot, Xu, and Young [PXY18], and denoted $Z_{\mathsf{M}}(x)$. One of the main contributions in [BHM+22] consists of interpreting these three polynomials in the following way:

- $P_{\mathsf{M}}(x)$ is the Hilbert series of the stalk of the intersection cohomology module of M at the empty set.
- $Z_{\mathsf{M}}(x)$ is the Hilbert series of the intersection cohomology module of M.
- $Q_{\mathsf{M}}(x)$ is the Hilbert series of the trivial graded module in the Rouquier complex of M.

The three interpretations above imply that these three polynomials appearing have nonnegative coefficients for every matroid M. However, computing the above polynomials is often a daunting task even for small matroids.

In [BV20], Braden and Vysogorets presented a formula that relates the Kazhdan–Lusztig polynomial of a matroid M to that of the matroid obtained by deleting an element i, denoted $\mathsf{M} \setminus i$, as well as various restrictions and contractions of M. Specifically, for a simple matroid M where i is not a coloop, their main result states:

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Theorem 1.1 ([BV20, Theorem 2.8]) *Let* M *be a loopless matroid of rank* k*, and let* $i \in E$ *be an element of the ground set that is not a coloop. Then,*

$$P_{\mathsf{M}}(x) = P_{\mathsf{M}\smallsetminus\{i\}}(x) - xP_{\mathsf{M}/\{i\}}(x) + \sum_{F\in\mathcal{S}_i} \tau\left(\mathsf{M}/(F\cup\{i\})\right) x^{\frac{k-\mathrm{rk}(F)}{2}} P_{\mathsf{M}|_F}(x),$$

$$Z_{\mathsf{M}}(x) = Z_{\mathsf{M}\smallsetminus\{i\}}(x) + \sum_{F\in\mathcal{S}_i} \tau\left(\mathsf{M}/(F\cup\{i\})\right) x^{\frac{k-\mathrm{rk}(F)}{2}} Z_{\mathsf{M}|_F}(x).$$

In the above statement, the notation $\underline{\mathcal{S}}_i$ stands for a distinguished set of flats:

$$\underline{\mathcal{S}}_i = \underline{\mathcal{S}}_i(\mathsf{M}) = \{F \in \mathcal{L}(\mathsf{M}) : \varnothing \subsetneq F \subsetneq E \smallsetminus \{i\} \text{ and } F\cup\{i\} \in \mathcal{L}(\mathsf{M})\}, \tag{1}$$

whereas $\tau(\mathsf{M})$ denotes

$$\tau(\mathsf{M}) := \begin{cases} [x^{\frac{\mathrm{rk}(\mathsf{M})-1}{2}}]P_{\mathsf{M}}(x) & \text{if } \mathrm{rk}(\mathsf{M}) \text{ is odd,} \\ 0 & \text{if } \mathrm{rk}(\mathsf{M}) \text{ is even.} \end{cases}$$

Luis: We should introduce and expand about the relevance of $Y_{\mathsf{M}}(x)$ (see below).

1.1. **Main results.** Let us introduce a counterpart for the sets $\underline{\mathcal{S}}_i$ appearing in equation (1). We define

$$\mathcal{T}_i := \{F \in \mathcal{L}(\mathsf{M}) : i \in F \text{ and } F \setminus \{i\} \notin \mathcal{L}(\mathsf{M})\}. \tag{2}$$

Theorem 1.2 *Let* M *be a matroid on* E *and let* $i \in E$ *be an element that is not a coloop. Then, the following identity holds:*

$$Q_{\mathsf{M}}(x) = Q_{\mathsf{M}\smallsetminus i}(x) + (1+x)Q_{\mathsf{M}/i}(x) - \sum_{F\in\mathcal{T}_i} \tau(\mathsf{M}|_F/i)\, x^{\mathrm{rk}(F)/2}\, Q_{\mathsf{M}/F}(x).$$

We now introduce the Y-polynomial of a matroid M, which is defined as:

$$Y_{\mathsf{M}}(x) := (-1)^{\mathrm{rk}(\mathsf{M})} \sum_{F\in\mathcal{L}(\mathsf{M})} (-1)^{\mathrm{rk}(F)} x^{\mathrm{rk}(\mathsf{M})-\mathrm{rk}(F)} \mu[F,E] Q_{\mathsf{M}|_F}(x), \tag{3}$$

where $\mu[F,E]$ is the Möbius function of the interval $[F,E]$ in the lattice of flats $\mathcal{L}(\mathsf{M})$.

Theorem 1.3 *Let* M *be a matroid on* E *and let* $i \in E$ *be an element that is not a coloop. Then, the following identity holds:*

$$Y_{\mathsf{M}}(x) = Y_{\mathsf{M}\smallsetminus i}(x) + (1+x)Y_{\mathsf{M}/i}(x) - \sum_{F\in\mathcal{T}_i} \tau(\mathsf{M}|_F/i)\, x^{\mathrm{rk}(F)/2}\, Y_{\mathsf{M}/F}(x).$$

Remark 1.4 The deletion formulas for $Q_{\mathsf{M}}(x)$ and $Y_{\mathsf{M}}(x)$ look exactly the same. Compare this against the deletion formulas of $P_{\mathsf{M}}(x)$ and $Z_{\mathsf{M}}(x)$.

Corollary 1.5 *Let* $\mathsf{C}_{a,b}$ *be the graphic matroid that results from gluing two cycles of length* a *and* b *along an edge. Then,*

$$\begin{aligned} Q_{\mathsf{C}_{a,b}}(x) = {} & Q_{\mathsf{U}_{n-2,n-1}}(x) + (1+x)Q_{\mathsf{U}_{a-2,a-1}}(x)\, Q_{\mathsf{U}_{b-2,b-1}}(x) \\ & - \left(\tau(\mathsf{U}_{a-2,a-1})\, x^{\frac{a-1}{2}}\, Q_{\mathsf{U}_{b-2,b-1}}(x) + \tau(\mathsf{U}_{b-2,b-1})\, x^{\frac{b-1}{2}}\, Q_{\mathsf{U}_{a-2,a-1}}(x)\right). \end{aligned}$$

where $n = a + b - 1$.

Example 1.6 In Figure 1 we depicted the graph that induces the graphic matroid $\mathsf{C}_{4,6}$.

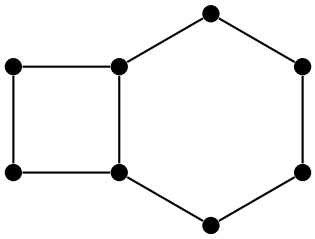


FIGURE 1. The graphic matroid $\mathsf{C}_{4,6}$.

Theorem 1.7 *Let* M *be a corank* 2 *matroid without coloops. Assume that for each* r, λ_r *denotes the number of stressed subsets of rank* r *and size* $r+1$ *in* M. *Then*

$$Q_{\mathsf{M}}(x) = Q_{\mathsf{U}_{n-2,n}}(x) - \sum_{r} \lambda_r \sum_{a=2}^{n-r-1} \left(Q_{\mathsf{C}_{a,n+1-a}}(x) - Q_{\mathsf{U}_{a-1,a}}(x)\, Q_{\mathsf{U}_{n-a-1,n-a}}(x) \right),$$

$$Y_{\mathsf{M}}(x) = Y_{\mathsf{U}_{n-2,n}}(x) - \sum_{r} \lambda_r \sum_{a=2}^{n-r-1} \left(Y_{\mathsf{C}_{a,n+1-a}}(x) - Y_{\mathsf{U}_{a-1,a}}(x)\, Y_{\mathsf{U}_{n-a-1,n-a}}(x) \right).$$

In particular, if M *arises from a partition* λ *of* $[n]$, *then*

$$Q_{\mathsf{M}}(x) = Q_{\mathsf{U}_{n-2,n}}(x) - \sum_{\substack{S \in \lambda \\ |S| \geq 2}} \sum_{a=2}^{|S|} \left(Q_{\mathsf{C}_{a,n+1-a}}(x) - Q_{\mathsf{U}_{a-1,a}}(x)\, Q_{\mathsf{U}_{n-a-1,n-a}}(x) \right),$$

$$Y_{\mathsf{M}}(x) = Y_{\mathsf{U}_{n-2,n}}(x) - \sum_{\substack{S \in \lambda \\ |S| \geq 2}} \sum_{a=2}^{|S|} \left(Y_{\mathsf{C}_{a,n+1-a}}(x) - Y_{\mathsf{U}_{a-1,a}}(x)\, Y_{\mathsf{U}_{n-a-1,n-a}}(x) \right).$$

Theorem 1.8 *There is a counterexample to Conjecture ? which consists of a matroid of rank* 19 *on* 21 *elements.*

2. PROOF OF THE DELETION FORMULAS

Let M be a matroid and $\mathcal{L}(\mathsf{M})$ be its lattice of flats. Following [BV20], we define the module $\mathcal{H}(\mathsf{M})$ of M as the free $\mathbb{Z}[x, x^{-1}]$-module generated by the elements of $\mathcal{L}(\mathsf{M})$. Elements of $\mathcal{H}(\mathsf{M})$ are formal sums of the form

$$\alpha = \sum_{F \in \mathcal{L}(\mathsf{M})} \alpha_F \cdot F, \quad \alpha_F \in \mathbb{Z}[x, x^{-1}].$$

The module $\mathcal{H}(\mathsf{M})$ admits the Kazhdan–Lusztig basis ζ^F for $F \in \mathcal{L}(\mathsf{M})$ defined by

$$\zeta^F = \sum_{G \leq F} x^{\mathrm{rk}(F)-\mathrm{rk}(G)} P_{\mathsf{M}^F_G}(x^{-2}) \cdot G.$$

where $P_{\mathsf{M}^F_G}(x)$ is the Kazhdan–Lusztig polynomial of the matroid M^F_G.

Let $\hat{Q}_{\mathsf{M}}(x)$ denote the inverse of the Kazhdan–Lusztig polynomial, $P_{\mathsf{M}}(x)$, of M with respect to the incidence algebra of $\mathcal{L}(\mathsf{M})$. Then, we have

$$Q_{\mathsf{M}}(x) = (-1)^{\mathrm{rk}(\mathsf{M})} \hat{Q}_{\mathsf{M}}(x). \tag{4}$$

Lemma 2.1 *The standard basis* $\{F\}_{F \in \mathcal{L}(\mathsf{M})}$ *of* $\mathcal{H}(\mathsf{M})$ *satisfies*

$$F = \sum_{G \leq F} t^{\mathrm{rk}(F)-\mathrm{rk}(G)} \hat{Q}_{\mathsf{M}^F_G}(t^{-2}) \cdot \zeta^G. \tag{5}$$

2.1. **Homomorphism $\Delta : \mathcal{H}(\mathsf{M}) \to \mathcal{H}(\mathsf{M} \setminus i)$.** Let i be an element of the ground set of M that is not a coloop. There is a surjective map $\mathcal{L}(\mathsf{M}) \to \mathcal{L}(\mathsf{M} \setminus i)$ sending a flat $F \in \mathcal{L}(\mathsf{M})$ to $F \setminus i \in \mathcal{L}(\mathsf{M} \setminus i)$. There also exists a module homomorphism $\Delta : \mathcal{H}(\mathsf{M}) \to \mathcal{H}(\mathsf{M} \setminus i)$ which is $\mathbb{Z}[x, x^{-1}]$-linear and is defined on the basis elements $F \in \mathcal{L}(\mathsf{M})$ by $\Delta(F) = x^{\mathrm{rk}_{\mathsf{M}\setminus i}(F\setminus i) - \mathrm{rk}_{\mathsf{M}}(F)} \cdot (F \setminus i)$. In particular, we have

$$\Delta(F) = \begin{cases} F & \text{if } F \not\ni i \\ F \setminus i & \text{if } F \ni i \text{ and } F \setminus i \notin \mathcal{L}(\mathsf{M}) \\ x^{-1} \cdot (F \setminus i) & \text{if } F \ni i \text{ and } F \setminus i \in \mathcal{L}(\mathsf{M}). \end{cases} \tag{6}$$

This definition is taken from Section 2.6 of [BV20].

Lemma 2.2 *Let $F \in \mathcal{L}(\mathsf{M})$ be a flat such that $i \in F$ and $F \setminus i \notin \mathcal{L}(\mathsf{M})$. Then*

$$\Delta(\zeta^F) = \zeta^{F\setminus i} + \sum_{G \in \underline{\mathcal{S}}_i(\mathsf{M}^F)} \tau(\mathsf{M}^F_{G\cup i}) \cdot \zeta^G \tag{7}$$

where $\underline{\mathcal{S}}_i(\mathsf{M}^F) = \{G \in \mathcal{L}(\mathsf{M}^F) \mid i \notin G \text{ and } G \cup i \in \mathcal{L}(\mathsf{M}^F)\}$.

Proof. Since i is not a coloop of the matroid M^F, the formula in equation 12 of [BV20] applies to the matroid M^F and the element i. □

Lemma 2.3 *Let $F \in \mathcal{L}(\mathsf{M})$ be a flat such that $i \in F$ and $F \setminus i \in \mathcal{L}(\mathsf{M})$, then*

$$\Delta(\zeta^F) = (x + x^{-1})\zeta^{F\setminus i}$$

where $\zeta^{F\setminus i}$ on the right-hand side is the ζ-element in $\mathcal{H}(\mathsf{M} \setminus i)$ associated with the flat $F \setminus i \in \mathcal{L}(\mathsf{M} \setminus i)$.

Proof. Let $F_0 = F \setminus i$. We are given $F_0 \in \mathcal{L}(\mathsf{M})$. By the definition of $\Delta(G)$:

- If $G \leq F$ and $i \notin G$: Then $G \leq F_0 \in \mathcal{L}(\mathsf{M} \setminus i)$. Since $G \in \mathcal{L}(\mathsf{M})$ and $i \notin G$, G is also a flat in $M \setminus i$, so $G \in \mathcal{L}(\mathsf{M} \setminus i)$. In this case, $\Delta(G) = G$.
- If $G \leq F$ and $i \in G$: Let $G_0 = G \setminus i$. Then $G_0 \leq F_0 \in \mathcal{L}(\mathsf{M} \setminus i)$. Since $F_0 \in \mathcal{L}(\mathsf{M})$, it follows that $G_0 = G \cap F_0 \in \mathcal{L}(\mathsf{M})$. In this case, $\Delta(G) = x^{-1}G_0$.

Applying this to $\Delta(\zeta^F)$:

$$\begin{aligned}
\Delta(\zeta^F) &= \Delta\left(\sum_{G\leq F} x^{\mathrm{rk}_{\mathsf{M}}(F)-\mathrm{rk}_{\mathsf{M}}(G)} P_{\mathsf{M}^F_G}(x^{-2}) \cdot G\right) \\
&= \sum_{G\leq F, G\not\ni i} x^{\mathrm{rk}_{\mathsf{M}}(F)-\mathrm{rk}_{\mathsf{M}}(G)} P_{\mathsf{M}^F_G}(x^{-2}) \cdot \Delta(G) \\
&\quad + \sum_{G\leq F, G\ni i} x^{\mathrm{rk}_{\mathsf{M}}(F)-\mathrm{rk}_{\mathsf{M}}(G)} P_{\mathsf{M}^F_G}(x^{-2}) \cdot \Delta(G) \\
&= \sum_{G\leq F_0} x^{\mathrm{rk}_{\mathsf{M}}(F)-\mathrm{rk}_{\mathsf{M}}(G)} P_{\mathsf{M}^F_G}(x^{-2}) \cdot G \\
&\quad + \sum_{\substack{G_0\leq F_0 \\ (G=G_0\cup\{i\})}} x^{\mathrm{rk}_{\mathsf{M}}(F)-\mathrm{rk}_{\mathsf{M}}(G_0\cup\{i\})} P_{\mathsf{M}^F_{G_0\cup\{i\}}}(x^{-2}) \cdot (x^{-1}G_0).
\end{aligned}$$

We use the rank relations:

- $\mathrm{rk}_{\mathsf{M}}(F) = \mathrm{rk}_{\mathsf{M}\setminus i}(F_0) + 1$.
- For $G \leq F_0$: $\mathrm{rk}_{\mathsf{M}}(G) = \mathrm{rk}_{\mathsf{M}\setminus i}(G)$.
- For $G_0 \leq F_0$: $\mathrm{rk}_{\mathsf{M}}(G_0 \cup \{i\}) = \mathrm{rk}_{\mathsf{M}\setminus i}(G_0) + 1$.

And the Kazhdan-Lusztig polynomial identities under these conditions:

- For $G \leq F_0$: $\mathsf{M}_G^F \cong (\mathsf{M} \setminus i)_G^{F_0} \oplus U_{1,1}(\{i\})$, so $P_{\mathsf{M}_G^F}(x^{-2}) = P_{(\mathsf{M}\setminus i)_G^{F_0}}(x^{-2})$.
- For $G_0 \leq F_0$: $\mathsf{M}_{G_0\cup\{i\}}^F \cong (\mathsf{M} \setminus i)_{G_0}^{F_0}$, so $P_{\mathsf{M}_{G_0\cup\{i\}}^F}(x^{-2}) = P_{(\mathsf{M}\setminus i)_{G_0}^{F_0}}(x^{-2})$.

Substituting these into the sums:

$$\begin{aligned}
\Delta(\zeta^F) &= \sum_{G\leq F_0} x^{(\mathrm{rk}_{\mathsf{M}\setminus i}(F_0)+1)-\mathrm{rk}_{\mathsf{M}\setminus i}(G)} P_{(\mathsf{M}\setminus i)_G^{F_0}}(x^{-2})\cdot G \\
&\quad + x^{-1}\sum_{G_0\leq F_0} x^{(\mathrm{rk}_{\mathsf{M}\setminus i}(F_0)+1)-(\mathrm{rk}_{\mathsf{M}\setminus i}(G_0)+1)} P_{(\mathsf{M}\setminus i)_{G_0}^{F_0}}(x^{-2})\cdot G_0 \\
&= x\sum_{G\leq F_0, G\in\mathcal{L}(\mathsf{M}\setminus i)} x^{\mathrm{rk}_{\mathsf{M}\setminus i}(F_0)-\mathrm{rk}_{\mathsf{M}\setminus i}(G)} P_{(\mathsf{M}\setminus i)_G^{F_0}}(x^{-2})\cdot G \\
&\quad + x^{-1}\sum_{G_0\leq F_0, G_0\in\mathcal{L}(\mathsf{M}\setminus i)} x^{\mathrm{rk}_{\mathsf{M}\setminus i}(F_0)-\mathrm{rk}_{\mathsf{M}\setminus i}(G_0)} P_{(\mathsf{M}\setminus i)_{G_0}^{F_0}}(x^{-2})\cdot G_0.
\end{aligned}$$

Using the definition of $\zeta_{M\setminus i}^{F_0}$:

$$\begin{aligned}
\Delta(\zeta^F) &= x\cdot\zeta_{M\setminus i}^{F_0} + x^{-1}\cdot\zeta_{M\setminus i}^{F_0} \\
&= (x+x^{-1})\zeta_{M\setminus i}^{F_0}.
\end{aligned}$$

□

Proof of 1.2. Applying the homomorphism Δ to equation 5 when $F = E$, we get

$$E\setminus e = \sum_{G\leq E} x^{\mathrm{rk}_{\mathsf{M}}(E)-\mathrm{rk}_{\mathsf{M}}(G)}\hat{Q}_{\mathsf{M}_G^E}(x^{-2})\cdot\Delta(\zeta^G).$$

Using lemma 2.1, the coefficient of $\zeta^\emptyset$ in the left-hand side is $x^{\mathrm{rk}_{\mathsf{M}\setminus i}(E\setminus e)-\mathrm{rk}_{\mathsf{M}\setminus i}(\emptyset)}\cdot\hat{Q}_{\mathsf{M}_\emptyset^{E\setminus e}}(x^{-2}) = x^{\mathrm{rk}_{\mathsf{M}}(M)}\cdot\hat{Q}_{\mathsf{M}\setminus e}(x^{-2})$.

By Lemma 2.2 and 2.3, the coefficient $\zeta^\emptyset$ on the right-hand side is non-zero only when $G\in\{\emptyset, e\}\cup\mathcal{T}_i$. In each of these cases, we have the following:

- For $G=\emptyset$, we have $\Delta(\zeta^\emptyset) = \zeta^\emptyset$.
- For $G = e$, we have $\Delta(\zeta^e) = (x+x^{-1})\zeta^\emptyset$.
- For $G\in\mathcal{T}_i$, we have $\Delta(\zeta^G) = \tau(\mathsf{M}_i^G)\cdot\zeta^\emptyset +$ terms not including $\zeta^\emptyset$.

Collecting the coefficients of $\zeta^\emptyset$ on the right-hand side, we have:

$$\begin{aligned}
&x^{\mathrm{rk}_{\mathsf{M}}(E)-\mathrm{rk}_{\mathsf{M}}(\emptyset)}\hat{Q}_{\mathsf{M}_\emptyset^E}(x^{-2}) + x^{\mathrm{rk}_{\mathsf{M}}(E)-\mathrm{rk}_{\mathsf{M}}(e)}(x+x^{-1})\hat{Q}_{\mathsf{M}_e^E}(x^{-2}) \\
&\qquad + \sum_{G\in\mathcal{T}_i} x^{\mathrm{rk}_{\mathsf{M}}(E)-\mathrm{rk}_{\mathsf{M}}(G)}\hat{Q}_{\mathsf{M}_G^E}(x^{-2})\cdot\tau(\mathsf{M}_e^G) \\
&= x^{\mathrm{rk}_{\mathsf{M}}(M)}\hat{Q}_{\mathsf{M}}(x^{-2}) + x^{\mathrm{rk}_{\mathsf{M}}(M)-1}(x+x^{-1})\hat{Q}_{\mathsf{M}_e}(x^{-2}) \\
&\qquad + \sum_{G\in\mathcal{T}_i} x^{\mathrm{rk}_{\mathsf{M}}(M)-\mathrm{rk}_{\mathsf{M}}(G)}\hat{Q}_{\mathsf{M}_G}(x^{-2})\cdot\tau(\mathsf{M}_e^G).
\end{aligned}$$

Equating the coefficients of $\zeta^\emptyset$ on both sides, we obtain:

$$\hat{Q}_{\mathsf{M}\setminus e}(x^{-2}) = \hat{Q}_{\mathsf{M}}(x^{-2}) + (1+x^{-2})\hat{Q}_{\mathsf{M}_e}(x^{-2}) + \sum_{G\in\mathcal{T}_i} x^{-\mathrm{rk}_{\mathsf{M}}(G)}\hat{Q}_{\mathsf{M}_G}(x^{-2})\cdot\tau(\mathsf{M}_e^G).$$

Taking $x^{-2} = t$ and rearranging yields:

$$\hat{Q}_{\mathsf{M}}(t) = \hat{Q}_{\mathsf{M}\setminus i}(t) - (1+t)\cdot\hat{Q}_{\mathsf{M}_i}(t) - \sum_{G\in\mathcal{T}_i}\tau(\mathsf{M}_i^G)\cdot t^{\mathrm{rk}_{\mathsf{M}}(G)/2}\cdot\hat{Q}_{\mathsf{M}_G}(t). \tag{8}$$

Finally, using the substitution $\hat{Q}_{\mathsf{M}}(t) = (-1)^{\mathrm{rk}(M)} Q_{\mathsf{M}}(t)$, we obtain:

$$Q_{\mathsf{M}}(t) = Q_{\mathsf{M}\setminus i}(t) + (1+t) Q_{\mathsf{M}_i}(t) - \sum_{F \in \mathcal{T}_i} \tau(\mathsf{M}_i^F)\, t^{\mathrm{rk}(F)/2}\, Q_{\mathsf{M}_F}(t)$$

as desired. □

Example 2.4 Let $M := U_{m,d}$ be the uniform matroid of rank d on $m+d$ elements with $d \geq 1$. Then

$$Q_{U_{m,d}}(x) = Q_{U_{m-1,d}}(x) + (1+x) \cdot Q_{U_{m,d-1}}(x) - \tau(U_{m,d-1}) \cdot x^{d/2}.$$

Proof. As none of the elements in the ground set are coloops, we may apply the deletion formula for a generic element e in the ground set. The set $\mathcal{T}_e$ then consists only of the top flat E and the sum over $\mathcal{T}_e$ reduces to

$$\tau(M_e) \cdot x^{\mathrm{rk}_{\mathsf{M}}(M)/2} = \tau(U_{m,d-1}) \cdot x^{d/2}.$$

□

Example 2.5 Let M be the parallel connection of the cycle matroids $U_{1,m}$ and $U_{1,n}$ along the element i. Then

$$\begin{aligned} Q_{\mathsf{M}}(x) = {} & Q_{U_{1,m+n-1}}(x) + (1+x) \cdot Q_{U_{1,m-1}}(x) \cdot Q_{U_{1,n-1}}(x) \\ & - \left[\tau(U_{1,m-1}) \cdot x^{m/2} Q_{U_{1,n-1}}(x) + \tau(U_{1,n-1}) \cdot x^{n/2} Q_{U_{1,m-1}}(x) \right]. \end{aligned}$$

Specifically, when the ranks m and n are odd, the τ terms are 0 and the formula simplifies to:

$$Q_{\mathsf{M}}(x) = Q_{U_{1,m+n-1}}(x) + (1+x) \cdot Q_{U_{1,m-1}}(x) \cdot Q_{U_{1,n-1}}(x).$$

Proof. We first note that for the parallel connection M of $U_{1,m}$ and $U_{1,n}$ along the element i, the set $\mathcal{T}_i$ consists of the top flat E and the two flats $F \cong U_{1,m}$ and $G \cong U_{1,n}$. Thus the sum over $\mathcal{T}_i$ reduces to:

$$\begin{aligned} & \tau(\mathsf{M}_i^E) \cdot x^{\mathrm{rk}(E)/2} Q_{\mathsf{M}_E}(x) + \tau(\mathsf{M}_i^F) \cdot x^{\mathrm{rk}(F)/2} Q_{\mathsf{M}_F}(x) + \tau(\mathsf{M}_i^G) \cdot x^{\mathrm{rk}(G)/2} Q_{\mathsf{M}_G}(x) \\ = {} & \tau(\mathsf{M}_i^E) \cdot x^{(m+n-1)/2} + \tau(U_{1,m-1}) \cdot x^{m/2} Q_{U_{1,n-1}}(x) + \tau(U_{1,n-1}) \cdot x^{n/2} Q_{U_{1,m-1}}(x). \end{aligned}$$

Since $\mathsf{M}_i^E \cong U_{1,m-1} \oplus U_{1,n-1}$, we have $\tau(\mathsf{M}_i^E) = 0$ by Lemma 2.7 in [BV20]. Using $Q_{\mathsf{M}\setminus i}(x) = Q_{U_{1,m+n-1}}(x)$ and $Q_{\mathsf{M}_i}(x) = Q_{U_{1,m-1} \oplus U_{1,n-1}}(x) = Q_{U_{1,m-1}}(x) \cdot Q_{U_{1,n-1}}(x)$, we obtain:

$$\begin{aligned} Q_{\mathsf{M}}(x) = {} & Q_{U_{1,m+n-1}}(x) + (1+x) \cdot Q_{U_{1,m-1}}(x) \cdot Q_{U_{1,n-1}}(x) \\ & - \left[\tau(U_{1,m-1}) \cdot x^{m/2} Q_{U_{1,n-1}}(x) + \tau(U_{1,n-1}) \cdot x^{n/2} Q_{U_{1,m-1}}(x) \right]. \end{aligned}$$

□

Example 2.6 Let $\mathsf{M} = PG(r-1, q)$ be the projective geometry of rank r over the finite field $GF(q)$, for $r \geq 2$. Then the deletion formula for its inverse Kazhdan-Lusztig polynomial is:

$$Q_{\mathsf{M}}(x) = Q_{\mathsf{M}\setminus e}(x) + (1+x) \cdot Q_{PG(r-2,q)}(x) - \begin{bmatrix} r-1 \\ 1 \end{bmatrix}_q \cdot x \cdot Q_{PG(r-3,q)}(x)$$

where $\begin{bmatrix} r-1 \\ 1 \end{bmatrix}_q = \frac{q^{r-1}-1}{q-1}$ is the number of lines through the point e in M.

Proof. We start from the formula for $Q_{\mathsf{M}}(x)$ in Theorem 1.2 and substitute $\mathsf{M} = PG(r-1,q)$. The lattice of flats of a projective geometry is modular. For any matroid N with a modular lattice of flats, its Kazhdan-Lusztig polynomial $P_{\mathsf{N}}(t)$ is 1 by [EPW16][Proposition 2.14]. The coefficient $\tau(\mathsf{N})$ is the coefficient of $t^{(\mathrm{rk}(\mathsf{N})-1)/2}$ in $P_{\mathsf{N}}(t)$, which is zero if $\mathrm{rk}(\mathsf{N}) \geq 2$.

The sum in Theorem 1.2 is over the set $\mathcal{T}_e = \{G \in \mathcal{L}(\mathsf{M}) \mid e \in G \text{ and } G \setminus e \notin \mathcal{L}(\mathsf{M})\}$. For the projective geometry M, a flat G containing e is a linear subspace. If $\mathrm{rk}(G) \geq 2$, removing the point e results in a set $G \setminus e$ that is not a subspace, and thus not a flat. If $\mathrm{rk}(G) = 1$, then $G = \{e\}$ and $G \setminus e = \emptyset$, which is a flat. Thus, for $\mathsf{M} = PG(r-1,q)$, the set $\mathcal{T}_e$ consists of all flats containing e of rank 2 or greater.

For each $G \in \mathcal{T}_e$, the term in the sum is $\tau(\mathsf{M}_e^G) \cdot x^{\mathrm{rk}(\mathsf{M}_e^G)/2} \cdot Q_{\mathsf{M}_G}(x)$. The matroid M_e^G is the interval $[e, G]$ in $\mathcal{L}(\mathsf{M})$ isomorphic to $PG(k-2,q)$ where $k = \mathrm{rk}(G)$. The rank of M_e^G is $k-1$ and $\tau(\mathsf{M}_e^G)$, therefore, is non-zero only if the rank $k-1 = 1$, which implies $k = 2$.

Thus, the sum is over the set of rank-2 flats containing the point e. The number of such flats is the number of 2-dimensional subspaces of $V(r,q)$ containing a given 1-dimensional subspace, which is equal to the number of 1-dimensional subspaces in the quotient space $V(r,q)/\langle e\rangle \cong V(r-1,q)$. This number is $\begin{bmatrix} r-1 \\ 1 \end{bmatrix}_q$. For each such rank-2 flat G:

- $\tau(\mathsf{M}_e^G) = \tau(PG(0,q)) = 1$ since $PG(0,q)$ has rank 1.
- $Q_{\mathsf{M}_G}(x) = Q_{PG(r-3,q)}(x)$, since the contraction M_G of $\mathsf{M} = PG(r-1,q)$ by a rank-2 flat G is isomorphic to $PG(r-3,q)$.

The entire sum therefore reduces to:

$$\sum_{G \in \mathcal{T}_e, \mathrm{rk}(G)=2} \tau(\mathsf{M}_e^G) \cdot x^{\mathrm{rk}(G)/2} \cdot Q_{\mathsf{M}_G}(x) = \begin{bmatrix} r-1 \\ 1 \end{bmatrix}_q \cdot x \cdot Q_{PG(r-3,q)}(x).$$

Substituting this and the fact that the contraction M_e is isomorphic to $PG(r-2,q)$ back into the general formula from Theorem 1.2, we get:

$$Q_{PG(r-1,q)}(x) = Q_{M\setminus e}(x) + (1+x) \cdot Q_{PG(r-2,q)}(x) - \begin{bmatrix} r-1 \\ 1 \end{bmatrix}_q \cdot x \cdot Q_{PG(r-3,q)}(x). \quad \square$$

$\square$

Remark 2.7 The inverse Kazhdan-Lusztig polynomial of a projective geometry $PG(k-1,q)$ of rank k is known to be the constant polynomial $q^{\binom{k}{2}}$. This allows for substituting the known values for the Q polynomials in the formula to obtain an explicit expression for $Q_{\mathsf{M}\setminus e}(x)$. Rearranging the terms and substituting gives:

$$\begin{aligned} Q_{\mathsf{M}\setminus e}(x) &= Q_{PG(r-1,q)}(x) - (1+x)Q_{PG(r-2,q)}(x) + \begin{bmatrix} r-1 \\ 1 \end{bmatrix}_q x\, Q_{PG(r-3,q)}(x) \\ &= q^{\binom{r}{2}} - (1+x)q^{\binom{r-1}{2}} + \frac{q^{r-1}-1}{q-1} q^{\binom{r-2}{2}} \cdot x \\ &= q^{\binom{r}{2}} - q^{\binom{r-1}{2}} + \left(\frac{q^{r-1}-1}{q-1} q^{\binom{r-2}{2}} - q^{\binom{r-1}{2}} \right) \cdot x. \end{aligned}$$

2.2. **Deletion formula for $Y_{\mathsf{M}}(x)$.** In order to prove the deletion formula for $Y_{\mathsf{M}}(x)$, we consider the related polynomial $\hat{Y}_{\mathsf{M}}(x)$ given by:

$$\hat{Y}_{\mathsf{M}}(x) = \sum_{F \in \mathcal{L}(\mathsf{M})} \left(x^{\mathrm{crk}(F)} \mu[F,E] \right) \cdot \hat{Q}_{\mathsf{M}^F}(x).$$

We define $\mu^F = \sum_{G \leq F} x^{\mathrm{rk}(G)} \mu_{\mathsf{M}}[G, F] \cdot G$ as our new basis for $\mathscr{H}(\mathsf{M})$ and consider its image under the deletion map Δ. Here, $\mu_{\mathsf{M}}[G, F]$ is the Möbius function of the interval $[G, F]$ in the lattice of flats of M.

Lemma 2.8 *If e is not a coloop of M, then*

$$\Delta(\mu^E) = \mu^{E \setminus e}.$$

Proof. We have

$$\Delta(\mu^E) = \sum_{G \leq E} x^{\mathrm{rk}(G)} \mu[G, E] \cdot \Delta(G).$$

We partition the lattice of flats $\mathscr{L}(\mathsf{M})$ into four disjoint sets:

- $S = \{G \in \mathscr{L}(\mathsf{M}) \mid e \notin G \text{ and } G \cup e \in \mathscr{L}(\mathsf{M})\}$.
- $T = \{G \in \mathscr{L}(\mathsf{M}) \mid e \in G \text{ and } G \setminus e \in \mathscr{L}(\mathsf{M})\}$.
- $A = \{G \in \mathscr{L}(\mathsf{M}) \mid e \notin G \text{ and } G \cup e \notin \mathscr{L}(\mathsf{M})\}$.
- $B = \{G \in \mathscr{L}(\mathsf{M}) \mid e \in G \text{ and } G \setminus e \notin \mathscr{L}(\mathsf{M})\}$.

We can now rewrite the sum by splitting it over these four sets:

$$\begin{aligned}\Delta(\mu_E) = &\sum_{G \in S} x^{\mathrm{rk}(G)} \mu[G, E] \cdot \Delta(G) + \sum_{G \in T} x^{\mathrm{rk}(G)} \mu[G, E] \cdot \Delta(G) \\ &+ \sum_{G \in A} x^{\mathrm{rk}(G)} \mu[G, E] \cdot \Delta(G) + \sum_{G \in B} x^{\mathrm{rk}(G)} \mu[G, E] \cdot \Delta(G).\end{aligned}$$

Applying the definition of Δ to each sum:

- For $G \in S \cup A$ (flats not containing e), $\Delta(G) = G$.
- For $G \in T$, $\Delta(G) = x^{-1}(G \setminus e)$.
- For $G \in B$, $\Delta(G) = G \setminus e$.

Substituting these into the expression gives:

$$\begin{aligned}\Delta(\mu_E) = &\sum_{G \in S} x^{\mathrm{rk}(G)} \mu[G, E] \cdot G + \sum_{G \in T} x^{\mathrm{rk}(G)-1} \mu[G, E] \cdot (G \setminus e) \\ &+ \sum_{G \in A} x^{\mathrm{rk}(G)} \mu[G, E] \cdot G + \sum_{G \in B} x^{\mathrm{rk}(G)} \mu[G, E] \cdot (G \setminus e).\end{aligned}$$

There is a natural bijection between S and T where for each $F \in S$, we have $F \cup e \in T$. We re-index the sum over T using $F \in S$, which lets us combine the first two sums:

$$\begin{aligned}\Delta(\mu_E) = &\sum_{F \in S} \left(x^{\mathrm{rk}(F)} \mu[F, E] + x^{\mathrm{rk}(F \cup e)-1} \mu[F \cup e, E] \right) \cdot F \\ &+ \sum_{G \in A} x^{\mathrm{rk}(G)} \mu[G, E] \cdot G + \sum_{G \in B} x^{\mathrm{rk}(G)} \mu[G, E] \cdot (G \setminus e).\end{aligned}$$

For $F \in S$, $\mathrm{rk}(F \cup e) = \mathrm{rk}(F) + 1$. Further, since e is not a loop or coloop of the minor M/F, we have the standard deletion-contraction relation for the Möbius function:

$$\mu_{\mathsf{M}}[F, E] + \mu_{\mathsf{M}}[F \cup e, E] = \mu_{\mathsf{M} \setminus e}[F, E \setminus e].$$

We have analogous relations for the flats in the other sets. For a flat $G \in A$, the interval $[G, E]$ in $\mathscr{L}(\mathsf{M})$ is isomorphic to the interval $[G, E \setminus e]$ in $\mathscr{L}(\mathsf{M} \setminus e)$, which gives:

$$\mu_{\mathsf{M}}[G, E] = \mu_{\mathsf{M} \setminus e}[G, E \setminus e].$$

For a flat $G \in B$, its corresponding set $F = G \setminus e$ is a flat in $\mathsf{M} \setminus e$ but not in M. The intervals $[G, E]$ and $[G \setminus e, E \setminus e]$ are isomorphic, so the relationship between the Möbius functions in this case is:

$$\mu_{\mathsf{M}}[G, E] = \mu_{\mathsf{M}\setminus e}[G \setminus e, E \setminus e].$$

Substituting these identities into the expression for $\Delta(\mu^E)$ yields:

$$\begin{aligned}\Delta(\mu^E) = &\sum_{F\in S} x^{\mathrm{rk}(F)}\mu_{\mathsf{M}\setminus e}[F, E \setminus e] \cdot F \\ &+ \sum_{G\in A} x^{\mathrm{rk}(G)}\mu_{\mathsf{M}\setminus e}[G, E \setminus e] \cdot G + \sum_{G\in B} x^{\mathrm{rk}(G)}\mu_{\mathsf{M}\setminus e}[G \setminus e, E \setminus e] \cdot (G \setminus e).\end{aligned}$$

Since the terms on the right account for all the flats in $\mathcal{L}(\mathsf{M} \setminus e)$, we have:

$$\Delta(\mu^E) = \mu^{E\setminus e}.$$

□

Writing out μ^E in terms of Kazhdan-Lusztig basis elements, we have:

$$\begin{aligned}\mu^E &= \sum_{F\in\mathcal{L}(\mathsf{M})} x^{\mathrm{rk}(F)}\mu[F, E]\left(\sum_{G\le F} x^{\mathrm{rk}(F)-\mathrm{rk}(G)}\hat{Q}_{\mathsf{M}^F_G}(x^{-2}) \cdot \zeta^G\right) \\ &= \sum_{F\in\mathcal{L}(\mathsf{M})}\sum_{G\le F} x^{2\,\mathrm{rk}(F)-\mathrm{rk}(G)}\mu[F, E] \cdot \hat{Q}_{\mathsf{M}^F_G}(x^{-2}) \cdot \zeta^G \\ &= \sum_{G\in\mathcal{L}(\mathsf{M})}\sum_{F\ge G} x^{2\,\mathrm{rk}(F)-\mathrm{rk}(G)}\mu[F, E] \cdot \hat{Q}_{\mathsf{M}^F_G}(x^{-2}) \cdot \zeta^G \\ &= \sum_{G\in\mathcal{L}(\mathsf{M})} x^{2\,\mathrm{rk}(E)-\mathrm{rk}(G)}\left(\sum_{F\ge G} x^{2\,\mathrm{rk}(F)-2\,\mathrm{rk}(E)}\mu[F, E]\hat{Q}_{\mathsf{M}^F_G}(x^{-2})\right) \cdot \zeta^G \\ &= \sum_{G\in\mathcal{L}(\mathsf{M})} x^{2\,\mathrm{rk}(E)-\mathrm{rk}(G)}\hat{Y}_{\mathsf{M}_G}(x^{-2}) \cdot \zeta^G\end{aligned}$$

Applying the deletion map Δ to this expression and using the lemma, we get:

$$\begin{aligned}\Delta(\mu^E) &= \sum_{G\in\mathcal{L}(\mathsf{M})} x^{2\,\mathrm{rk}(E)-\mathrm{rk}(G)}\hat{Y}_{\mathsf{M}_G}(x^{-2}) \cdot \Delta(\zeta^G) \\ \text{or,}\qquad \mu^{E\setminus e} &= \sum_{G\in\mathcal{L}(\mathsf{M})} x^{2\,\mathrm{rk}(E\setminus e)-\mathrm{rk}(G)}\hat{Y}_{\mathsf{M}_G}(x^{-2}) \cdot \Delta(\zeta^G)\end{aligned}$$

The coefficient of $\zeta^\emptyset$ on the left is $x^{2\,\mathrm{rk}(E\setminus e)}\hat{Y}_{\mathsf{M}\setminus e}(x^{-2})$. As in the proof of Theorem **??**, the coefficient of $\zeta^\emptyset$ on the right is:

$$x^{2\,\mathrm{rk}(E)}\hat{Y}_{\mathsf{M}}(x^{-2}) - (x + x^{-1})x^{2\,\mathrm{rk}(E)-1}\hat{Y}_{\mathsf{M}_e}(x^{-2}) + \sum_{G\in S'} \tau(\mathsf{M}^G_e) \cdot x^{2\,\mathrm{rk}(E)-\mathrm{rk}(G)}\hat{Y}_{\mathsf{M}_G}(x^{-2}).$$

Equating the coefficients of $\zeta^\emptyset$ on both sides, we get:

$$\hat{Y}_{\mathsf{M}\setminus e}(x^{-2}) = \hat{Y}_{\mathsf{M}}(x^{-2}) - (1 + x^{-2})\hat{Y}_{\mathsf{M}_e}(x^{-2}) + \sum_{G\in S'} \tau(\mathsf{M}^G_e) \cdot x^{-\mathrm{rk}(G)}\hat{Y}_{\mathsf{M}_G}(x^{-2}).$$

Substituting $t = x^{-2}$, we get:

$$\hat{Y}_{\mathsf{M}\setminus e}(t) = \hat{Y}_{\mathsf{M}}(t) - (1 + t)\hat{Y}_{\mathsf{M}_e}(t) + \sum_{G\in S'} \tau(\mathsf{M}^G_e) \cdot x^{\mathrm{rk}(G)/2}\hat{Y}_{\mathsf{M}_G}(t).$$

Using the fact that $\hat{Y}_{\mathsf{M}}(t) = (-1)^{\mathrm{rk}(\mathsf{M})} Y_{\mathsf{M}}(t)$, we get the following formula:

Theorem 2.9 *Let* M *be a matroid and* e *be a non-coloop of* M*. Then,*

$$Y_{\mathsf{M}}(t) = Y_{\mathsf{M}\setminus e}(t) - (1+t)Y_{\mathsf{M}_e}(t) - \sum_{G \in S'} \tau(\mathsf{M}_e^G) \cdot x^{\mathrm{rk}(G)/2} Y_{\mathsf{M}_G}(t).$$

3. A counterexample to the real-rootedness conjecture

Let M be a matroid of rank 2 without loops. The lattice of flats of M has the empty set as its bottom element and E as its top element. Notice that this guarantees rank 1 flats of M form a partition of E. Conversely, any partition of E yields a loopless rank 2 matroid on E.

By taking duals, the above construction yields a procedure that builds a corank 2 matroid on E without coloops from every set partition of E.

Notice that rank 2 matroids without loops are trivially paving, and therefore, corank 2 matroids without coloops are copaving.

Example 3.1 Consider $n = 8$ and the partition of $\{1, \dots, 8\}$ consisting on the sets $\{1, 5, 6, 8\}$, $\{2\}$, $\{3, 4, 7\}$. The loopless rank 2 matroid that has these three sets as its rank 1 flats is in fact the graphic matroid induced by the graph depicted in Figure 2 (if there are more than three parts in the partition, these matroids fail to be graphic). The corank 2 matroid associated to this set partition is the corresponding matroid dual.

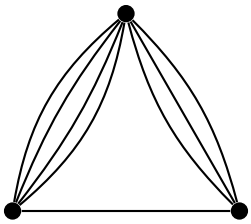


FIGURE 2. The edges of the left are labelled by $\{1, 5, 6, 8\}$, the edges on the right by $\{3, 4, 7\}$ and the bottom one by $\{2\}$.

Since the class of elementary split matroids contains all copaving matroids, and since the inverse Kazhdan–Lusztig polynomial is a valuative invariant, we may apply [FS24, Theorem 1.4] to compute $Q_{\mathsf{M}}(x)$ for any corank 2 matroid without coloops.

Theorem 3.2 *Let* M *be a corank* 2 *matroid without coloops. Assume that for each* r*,* λ_r *denotes the number of stressed subsets of rank* r *and size* $r+1$ *in* M*. Then*

$$Q_{\mathsf{M}}(x) = Q_{\mathsf{U}_{n-2,n}}(x) - \sum_r \lambda_r \sum_{a=2}^{n-r-1} \left(Q_{\mathsf{C}_{a,n+1-a}}(x) - Q_{\mathsf{U}_{a-1,a}}(x)\, Q_{\mathsf{U}_{n-a-1,n-a}}(x) \right),$$

$$Y_{\mathsf{M}}(x) = Y_{\mathsf{U}_{n-2,n}}(x) - \sum_r \lambda_r \sum_{a=2}^{n-r-1} \left(Y_{\mathsf{C}_{a,n+1-a}}(x) - Y_{\mathsf{U}_{a-1,a}}(x)\, Y_{\mathsf{U}_{n-a-1,n-a}}(x) \right).$$

In particular, if M *arises from a partition* λ *of* $[n]$*, then*

$$Q_{\mathsf{M}}(x) = Q_{\mathsf{U}_{n-2,n}}(x) - \sum_{\substack{S \in \lambda \\ |S| \geq 2}} \sum_{a=2}^{|S|} \left(Q_{\mathsf{C}_{a,n+1-a}}(x) - Q_{\mathsf{U}_{a-1,a}}(x)\, Q_{\mathsf{U}_{n-a-1,n-a}}(x) \right),$$

$$Y_{\mathsf{M}}(x) = Y_{\mathsf{U}_{n-2,n}}(x) - \sum_{\substack{S\in\lambda \\ |S|\geq 2}} \sum_{a=2}^{|S|} \left(Y_{\mathsf{C}_{a,n+1-a}}(x) - Y_{\mathsf{U}_{a-1,a}}(x)\, Y_{\mathsf{U}_{n-a-1,n-a}}(x)\right).$$

Proof. By a direct application of [FS24, Theorem 5.3], we obtain that for any coloopless corank 2 matroid on n elements, the following equality holds:

$$Q_{\mathsf{M}}(x) = Q_{\mathsf{U}_{n-2,n}}(x) - \sum_{r,h} \lambda_{r,h} \left(Q_{\Lambda_{r,n-2,h,n}}(x) - Q_{\mathsf{U}_{n-2-r,n-h}}(x) Q_{\mathsf{U}_{r,h}}(x)\right),$$

where $\lambda_{r,h}$ is the number of stressed subsets of M that have rank r and size h. Since M has corank 2, the definition of being a stressed subset implies that $h \in \{r, r+1, r+2\}$, and in fact the cases $h = r$ and $h = r+2$ cannot happen. Thus,

$$Q_{\mathsf{M}}(x) = Q_{\mathsf{U}_{n-2,n}}(x) - \sum_{h} \lambda_{r} \left(Q_{\Lambda_{r,n-2,r+1,n}}(x) - Q_{\mathsf{U}_{n-r-2,n-r-1}}(x) Q_{\mathsf{U}_{r,r+1}}(x)\right),$$

where now λ_r counts the number of stressed subsets of rank r and size $r+1$. By [FS24, Corollary 5.7], since the inverse Kazhdan–Lusztig polynomial is a valuative invariant, we obtain:

$$Q_{\Lambda_{r,n-2,r+1,n}}(x) = \sum_{a=2}^{n-r-1} Q_{\mathsf{C}_{a,n+1-a}}(x) - \sum_{a=2}^{n-r-2} Q_{\mathsf{U}_{a-1,a}}(x)\, Q_{\mathsf{U}_{n-a-1,n-a}}(x),$$

which yields:

$$Q_{\Lambda_{r,n-2,r+1,n}}(x) - Q_{\mathsf{U}_{n-r-2,n-r-1}}(x) Q_{\mathsf{U}_{r,r+1}}(x) = \sum_{a=2}^{n-r-1} \left(Q_{\mathsf{C}_{a,n+1-a}}(x) - Q_{\mathsf{U}_{a-1,a}}(x)\, Q_{\mathsf{U}_{n-a-1,n-a}}(x)\right).$$

Putting all the pieces together, we obtain:

$$Q_{\mathsf{M}}(x) = Q_{\mathsf{U}_{n-2,n}}(x) - \sum_{h} \lambda_{r} \sum_{a=2}^{n-r-1} \left(Q_{\mathsf{C}_{a,n+1-a}}(x) - Q_{\mathsf{U}_{a-1,a}}(x)\, Q_{\mathsf{U}_{n-a-1,n-a}}(x)\right),$$

as desired. □

Theorem 3.3 *The matroid induced by partitioning the set* $\{1, \ldots, 21\}$ *into six parts of sizes* $(4, 4, 4, 3, 3, 3)$ *is a counterexample to the real-rootedness conjecture of Xie–Zhang.*

Proof. Computing the inverse Kazhdan–Lusztig polynomial via the preceding theorem results in:

$$Q_{\mathsf{M}}(x) = 71162\,x^9 + 232662\,x^8 + 350404\,x^7 + 323646\,x^6 + 215169\,x^5 + 106659\,x^4 + 39217\,x^3 + 10323\,x^2 + 1790\,x + 163.$$

After taking the Hadamard product with $(1+x)^9$, we obtain:

$$(1+x)^9 * Q_{\mathsf{M}}(x) = 71162\,x^9 + 2093958\,x^8 + 12614544\,x^7 + 27186264\,x^6 + 27111294\,x^5 + 13439034\,x^4 + 3294228\,x^3 + 371628\,x^2 + 16110\,x + 163.$$

According to SageMath, this polynomial is not real-rooted. It possesses two complex conjugate zeros near $-1.0297 \pm 0.1097\,i$. □

Since $Q_{\mathsf{C}_{a,n+1-a}}(x)$ can be computed in terms of inverse KL polynomials of uniform matroids via the deletion formula, this gives a closed formula for any corank 2 matroid.

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